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1.1.1 | Summary

The design loads are based on the standards of the California Building Code (CBC) outlined below. Standard math symbols are profided for reference.

Table 1: California Building Standards

Standard	Organization	Description & Application
CBC Chapter 23	California Building Standards Commission	Governs the structural use of wood, including lumber, glulam, timber poles, and diaphragms.
CRC Chapters 5-8	California Building Standards Commission	Covers prescriptive construction requirements for one- and two-family light-frame wood structures.
AWC NDS	American Wood Council	Foundational national standard for ASD and LRFD design of timber structures.
AWC WFCM	American Wood Council	Design criteria for conventional light- frame wood construction to meet wind and seismic loads.
ANSI/APA PRG 320	APA â€“ The Engineered Wood Association	Standard for Performance-Rated CLT, essential for tall wood/mass timber buildings.
ICC 400	International Code Council	Design and construction standard for log structures.
ASTM Standards	ASTM International	Includes testing and evaluation methods such as ASTM D3957 (structural logs) and ASTM D7672 (rim boards).
AWPA U1 / M4	American Wood Protection Association	Standards for preservative treatment and use of wood poles, piles, and lumber in permanent installations.

Table 2: Math Abbreviations

Abbreviation	Definition
D	dead load
L	live load
D_m	module dead load
E	earthquake load
F_a	acceleration site coefficient
F_v	velocity site coefficient
F_N	normal wind force
GC_{Ms}	net moment static coefficient
GC_{Md}	net moment dynamic coefficient
GC_M	net moment coefficient

Abbreviation	Definition
GC_P	net pressure coefficient
GC_{Ps}	net static pressure coefficient
GC_{Pd}	net dynamic pressure coefficient
k_1	hazard coefficient
k_2	terrain and structure coefficient
k_3	topography coefficient
K_{zt}	topographic Factor
K_z	velocity pressure exposure coefficient
MRI	mean return interval
p_d	net design wind pressure on module - Pa
$SDOF$	single degree of freedom
S_s	short period mapped acceleration
$S_D S$	site design response acceleration
S_1	1 second period mapped acceleration
$S_M S$	short period parameter
S_{M1}	1 second period parameter
T	fundamental period of structure
M_{tor}	wind moment about panel center
T_0	short period spectral cap
T_S	long period spectral cap
V_b	basic wind speed
V_B	seismic design base shear
W	wind load / seismic weight of structure
F_b	NDS - Reference bending design value
F_t	NDS - Reference tension design value
F_v	NDS - Reference shear design value
F_{Cperp}	NDS - Reference compression perpendicular to grain
F_c	NDS - Reference compression parallel to grain
E	NDS - Modulus of elasticity
E_{min}	NDS - Reference modulus for stability and stiffness
G	NDS - Specific gravity
C_D	NDS - Load duration factor
C_M	NDS - Wet service factor
C_F	NDS - Size factor

Abbreviation	**Definition**
C_{fu}	NDS - Flat use factor
C_i	NDS - Incising factor
C_r	NDS - Repetitive member factor
C_t	NDS - Temperature factor
C_c	NDS - Curvature factor
C_l	NDS - Column stability factor
C_L	NDS - Beam stability factor
C_p	NDS - Column stability factor
C_b	NDS - Bearing area factor
C_T	NDS - Buckling stiffness factor
C_s	NDS - Slenderness factor
C_T	NDS - Buckling stiffness factor
F'_b	NDS - Adjusted bending design value
F'_t	NDS - Adjusted tension design value
F'_v	NDS - Adjusted shear design value
F'_{Cperp}	NDS - Adjusted compression perpendicular
F'_c	NDS - Adjusted compression parallel to grain
E'	NDS - Adjusted modulus of elasticity
E'_{min}	NDS - Adjusted modulus ofor stability and stiffness